



A novel image dehazing algorithm for complex natural environments

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ABSTRACT

As an essential preprocessing step in computer vision tasks, image dehazing is preparatory work in many practical applications. To ensure the navigation safety of intelligent ships in complex haze weather, this paper proposes a lightweight real-time dehazing method for the natural environment based on an all-in-one dehazing network. Specifically, a hybrid dilated convolution is used to construct a dehazing network, which effectively expands the receptive field and improves the ability to extract characteristic information without increasing the calculation amount of the model. Secondly, a hybrid attention module is developed to enhance the model's ability to remove haze while retaining more detailed features by giving weights to light haze and heavy haze areas in the image. Finally, considering the subjective perception of human eyes, a mixed loss function is designed to address image dimness and color distortion after dehazing. Through experiments on both synthetic and real-world haze datasets, the proposed method attains ideal dehazing results and has high real-time performance. It can dehaze under different haze conditions, address the critical challenge of degraded visual perception in severe weather conditions of intelligent ship navigation on inland rivers in complex environments, and promote intelligent ships' development.

1. Introduction

A typical natural phenomenon, haze, is formed by the strongly scattering medium in the atmosphere. In image acquisition, these suspended media interfere with the propagation of the reflected light of the scene in the imaging process, resulting in reduced brightness and dark color of the imaging results [1,2].

Currently, intelligent ships are developing rapidly and have gradually changed from experimental to practical. As an auxiliary means, computer vision perception has been widely used in ship navigation environment perception [3,4]. However, the environment of inland navigation is complex and changeable. Under the influence of haze, images or videos collected by shipborne vision sensors often have problems such as low visibility, poor contrast, dark background, and loss of detail information, which to some extent affect other intermediate and advanced vision tasks, including target detection [5], and tracking [6]. This problem is significant due to its practical relevance in enhancing safety and efficiency in maritime transportation, especially during severe weather conditions such as haze. Hazy weather significantly impairs visibility, posing risks to ship navigation and increasing

the likelihood of accidents. Therefore, effectively recovering the real-world scene from the haze image has become a critical problem in intelligent ship safety.

In recent years, image dehazing algorithms have been continuously advanced. These algorithms can be broadly categorized into three categories: image enhancement dehazing algorithm, image restoration dehazing algorithm, and deep learning-based dehazing algorithm. The image enhancement dehazing algorithm [7] enhances image quality by improving contrast and highlighting image features, without considering the physical model of image deterioration. This approach has low computational complexity and ensures real-time performance. The histogram equalization algorithm [8], based on the principle that the image has more prominence information when the gray level is evenly distributed, homogenizes the histogram to enhance the brightness and enlarge the details. However, it greatly influences contour information and easily causes distortion. Wu et al. [9] utilized the integer wavelet transform to distinguish the details and contour information of the image lossless and reversibly and more high-level information could be retained after image dehazing. Soni et al. [10] firstly used histogram equalization to enhance the image, then used the guided filter to recover

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the details lost after equalization, to make the dehazing image more natural. Retinex algorithm [11] possesses the benefits of color invariance and dynamic range compression and eliminates the effect of illumination by separating incident components to achieve the effect of image enhancement and dehazing. Li et al. [12] adopted an optimization function containing illumination and reflectance to obtain the image structure details under low illumination intensity and comprehensively considered image noise to improve the model's dehazing performance.

The atmospheric scattering model (ASM) primarily underlies the image restoration dehazing algorithm [13]. Through statistical analysis of a substantial quantity of images with haze and without haze, the mapping relationship between them is obtained, and then the inverse computation is performed according to the process of generating images with haze to obtain clear images. However, this kind of algorithm belongs to the ill-posed problem. The unknown quantities in the model's parameters, such as atmospheric light value, transmission, and image depth of field, are challenging to estimate accurately, affecting the image dehazing effect. To this end, researchers introduced a dehazing algorithm that utilizes prior conditions, among which the dark channel prior (DCP) algorithm is representative [14]. Xu et al. [15] proposed a fast dehazing algorithm that relies on fast bilateral filtering and DCP which solve the problem of the dark image after dehazing. Pan et al. [16] put forth a haze removal algorithm utilizing DCP and self-adaptive threshold. The evaluated atmospheric light value is used to solve the color distortion problem of the DCP algorithm after removing haze in the sky region.

As neural networks and deep learning continue to gain widespread use in computer vision, inspired by haze characteristics and the current dehazing algorithms, researchers have delved deeply into research on image dehazing that relies on convolutional neural networks (CNN) [17]. Cai et al. [18] established the DehazeNet model with a deep CNN structure. In this model, a new estimation of atmospheric degradation transmittance is carried out, and the bilateral rectifier linear element processes the atmospheric light value and transmittance to improve the image restoration quality. However, the constant value calculation of atmospheric light causes dehazing precision to decrease and the color distortion. To overcome the shortcomings of the DehazeNet model, Dudhane et al. [19] proposed a multi-scale network dehazing algorithm C2MSNet with context-aware characteristics. Firstly, color information is extracted from hazy images, and multi-channel depth maps are generated. Then, haze-free images are recovered from dark channel estimation scene transmission maps using CNN. Chao et al. [20] constructed a hybrid residual attention network designed to extract global and instance-level features from images, employing a pixel attention structure to fuse multi-scale features, resulting in satisfactory dehazing effects. Li et al. [21] put forth an efficient all-in-one dehazing network (AOD-Net), compared with the above model, AOD-Net transforms the ASM formula and directly learns from the image with haze to get the image without haze. The AOD-Net network structure is simple, only composed of five convolution layers, and can be well embedded in other advanced computer vision tasks. However, the single structure of the model also affects the learning ability of image features, which makes the dehazing effect of the haze area or distant area poor. For this reason, Qian et al. [22] proposed the FAOD-Net dehazing algorithm, which used lightweight depth-separable convolution to design network structures to advance the operation efficiency. Moreover, the pyramid pool module has been introduced to advance the capability of aggregation of different regional context information and improve the network's capacity to acquire global hazy image information. Yi et al. [23] suggested an image dehazing network with a double-cascaded end-to-end architecture for obtaining progressively clearer images from coarse to fine. Qin et al. [24] proposed a new feature attention module (FA) based on careful consideration of image channel and pixel features, which expanded the feature representation ability of CNN and could directly restore non-haze images from hazy images. Zhao et al. [25] designed a multi-scale global context awareness module to learn the long-distance

dependence between different features and achieve a good dehazing effect on the synthetic and real-world haze datasets. Liu et al. [26] proposed a three-stage dehazing network TSDNet, which introduced a multi-scale attention model to estimate the hazing component in the image and designed a double-branch extraction module and feature fusion model for haze features to obtain clear images. The model achieved a good dehazing effect in both synthetic and real-world imaging scenes. Ma et al. [27] employed hybrid feature convolution in an encoder-decoder structure to extract cross-scale feature information, significantly improving the clarity of hazy images. Ganguly et al. [28] first used the reconstructed atmospheric scattering model to estimate the atmospheric light value and transmittance and then designed a sparse haze model to estimate the haze characteristics. By efficiently cascading the two models, the haze removal of a single image was realized. Su et al. [29] combined the haze physical model with image translation technology, introduced sequential channel attention and spatial attention at the same time, and restored clear scenes through multi-scale fusion technology. Yin et al. [30] proposed an encoder-decoder dehaze network structure based on a pyramid pool and applied spatial and channel attention modules in parallel to the end of the encoder to capture the spatial information of haze images and the dependency relationship between channels. Experimental results show that this method can achieve a better dehazing effect. These studies highlight the importance of attention mechanisms and receptive field expansion in enhancing dehazing performance.

Currently, the dynamic Gaussian kernel algorithm shows significant potential in the field of image dehazing due to its smoothing filtering and noise suppression capabilities. Fan et al. [31] employed Gaussian process regression to establish a direct relationship between hazy images and depth-related transmission matrices. By learning local image priors, the model's dehazing ability is further enhanced, achieving effective visual restoration for both real-world and synthetic hazy images. Kumari et al. [32] combined an adaptive Gaussian notch filter with gamma correction to improve the accuracy of transmission depth feature estimation, effectively enhancing the quality of hazy images while preserving image edges and detail features. Sebastián et al. [33] proposed a dehazing method based on image morphology processing theory and Gaussian filter. This method removes hazy areas through morphological operations and smooths the image using a Gaussian filter, greatly reducing noise impact and achieving good dehazing results. The above studies indicate that dynamic Gaussian kernels have significant advantages in dehazing algorithms. However, current research still faces issues such as high computational complexity and strong dependency on parameter tuning, which present considerable limitations in practical applications.

1.1. Motivation and contributions

To address critical challenges in the field of computer vision and intelligent ship navigation in hazy weather conditions, this paper suggests an AOD-Net-based image dehazing algorithm. In this method, the physical process of haze formation is fully considered, and the dehazing performance of the model is enhanced by reconstructing the network structure and introducing the ECPA module and mixed loss function. Specifically, the main contributions of our method are summarized as follows:

- (1) Firstly, hybrid dilated convolution (HDC) is first introduced into the classical AOD-Net model to improve the size of the feature receptive field and minimize information loss, while keeping computation cost constant.
- (2) Secondly, a hybrid attention module (ECPA) is designed to enhance image fusion channel information and pixel information ability. By assigning different weights to the light haze and heavy haze regions in the image, the model can improve the image dehazing ability while retaining more details.

- (3) Finally, to address the issue of image dimness and color distortion after haze removal, the mixing loss function is designed for network training.
- (4) The empirical findings indicate that the algorithm suggested can attain a good haze removal outcome under different hazy scenarios, and a clearer and more natural inland river scene can be obtained, which offers technical assistance for the visual perception of intelligent ships and further promotes the development of intelligent ships.

The structure of this paper is organized as follows. We introduce related work on image dehazing in Section 1. In Section 2, we briefly introduce the mathematical model of ASM and the knowledge of classical AOD-Net networks. Specifically, we introduce the components of the algorithm proposed in detail in Section 3, including HDC, ECPA module, and mixed loss function. Section 4 compares our method with other state-of-the-art methods on different datasets. In Section 5, we discuss the importance of image dehazing in various real-world applications. Finally, we conclude the proposed method and discuss the future work in Section 6.

2. Principle of image dehazing

2.1. Introduction to ASM theory

ASM, as the classical physical description of hazy image generation, comprehensively considers the influence of haze on light scattering and refraction in the imaging process and is an essential theoretical basis of image dehazing technology, as depicted in Fig. 1.

Classic ASM [13] can be represented as:

$$I_{haze}(x) = J_{clean}(x)t_{trans}(x) + A(1 - t_{trans}(x)) \quad (1)$$

In Formula (1), the pixel in the image can be expressed as x ; the global atmospheric light value is A ; $I_{haze}(x)$ is the image with haze observed; $J_{clean}(x)$ is the corresponding image without haze; $t_{trans}(x)$ is the transmission relationship between the two, which can be expressed as:

$$t(x) = e^{-\beta d_{depth}(x)} \quad (2)$$

In Formula (2), β is the atmospheric scattering coefficient, and $d_{depth}(x)$ is the scene depth, that is, the gap between the target and the image acquisition device.

For Formula (1), it could be redesigned as:

$$J_{clean}(x) = \frac{1}{t_{trans}(x)} I_{haze}(x) - A \frac{1}{t_{trans}(x)} + A \quad (3)$$

According to Formula (3), assuming that A and $t_{trans}(x)$ can be accurately estimated from the detected hazy image, the matching clear image $J_{clean}(x)$ can be obtained.

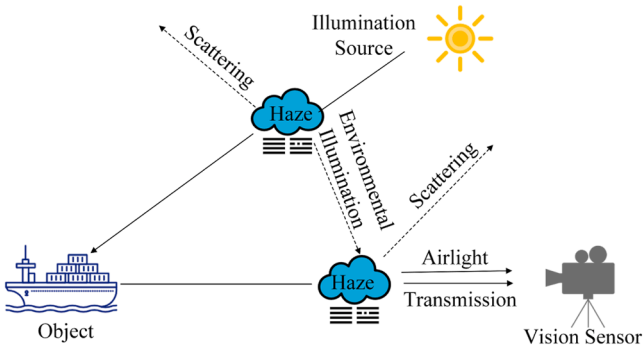


Fig. 1. Atmospheric scattering model in hazy imaging. It illustrates light scattering through haze from the illumination source to the vision sensor, causing reduced image contrast and distortion.

2.2. AOD-Net model

AOD-Net builds an end-to-end image dehazing model built upon CNN. By re-designing ASM, it can effectively avoid the issue of estimating a significantly high atmospheric light value A , which may be affected by the presence of white or sky areas, and learn clear images from hazy images to minimize reconstruction errors [21]. Specifically, by unifying variables $t_{trans}(x)$ and A of Formula (1) into parameter $K(x)$, simplified ASM can be expressed as:

$$J_{clean}(x) = K(x)I_{haze}(x) - K(x) + b \quad (4)$$

In Formula (4), b is a constant and default $b = 1$; $K(x)$ combines variables $t_{trans}(x)$ and A and is associated with the hazy image $I_{haze}(x)$ input by the model, thus:

$$K(x) = \frac{(I_{haze}(x) - A)/t_{trans}(x) + (A - b)}{I_{haze}(x) - 1} \quad (5)$$

Evidently, AOD-Net is made up of an $K(x)$ value estimation module and an image restoration module, which simplifies the traditional image dehazing problem into $K(x)$ value-solving process. The hazy images $I_{haze}(x)$ are inputted directly into the network, extract image feature information through the $K(x)$ estimation module, and recover clear images. The specific algorithm flow is depicted in Fig. 2.

$K(x)$ estimation module is the core structure of the AOD-Net model. It uses the network structure based on CNN to design a multi-scale convolution layer and splice and fuse image feature information to advance the capturing capability for information of different scales and make up for the information loss between features in the convolution process. The network structure is depicted in Fig. 3.

In Fig. 3, the $K(x)$ estimation module uses five convolution kernels of different sizes for convolutional operations to extract multi-scale feature information, and only three filters are used for each convolutional layer. Therefore, compared with other deep learning-based image dehazing models, the AOD-Net model is lighter. At the same time, Concat1 integrates the feature information of Conv1 and Conv2 layers and then further extracts the feature through Conv3 layers. Similarly, Concat2 fuses the feature information of Conv2 and Conv3 layers and is processed by Conv4 layers. Concat3 fuses the feature information of Conv1, Conv2, Conv3, and Conv4 convolution layers, and finally convolved by the Conv5 layer and extracts the feature information of different scales. Based on the above network structure, AOD-Net can obtain multi-scale feature information, and make up for the detail loss in the convolution procedure of each layer by splicing and blending features. By inputting a hazy image $I_{haze}(x)$ into the trained AOD-Net model, the haze-free image $J_{clean}(x)$ can be directly attained.

3. Proposed method

3.1. Hybrid dilated conv (HDC)

Inspired by CNN's feature extraction on different scales, expanding the receptive field can attain more feature information and improve network performance. However, in the traditional CNN structure, increasing the receptive field by increasing the network's depth or the convolution kernel's size leads to a dramatic increase in network parameters and computation. Therefore, dilated convolution introduces a dilated rate [34]. By adjusting this parameter, the receptive field can be expanded without advancing the amount of network computation to obtain multi-scale information. Nevertheless, after multiple dilated convolutions, there exists a discernible gap between each pixel, and the convolution results lack spatial correlation, consequently leading to the loss of local feature information.

To solve these problems, hybrid dilated convolution [35] is adopted in this paper to replace partial convolution operations in the AOD-Net model. Combining dilated convolution with different dilated rates

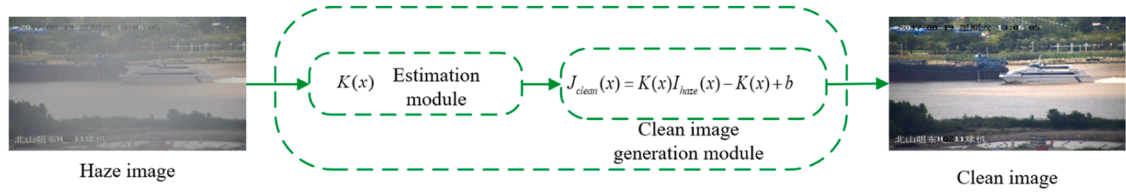


Fig. 2. AOD-Net algorithm flow.

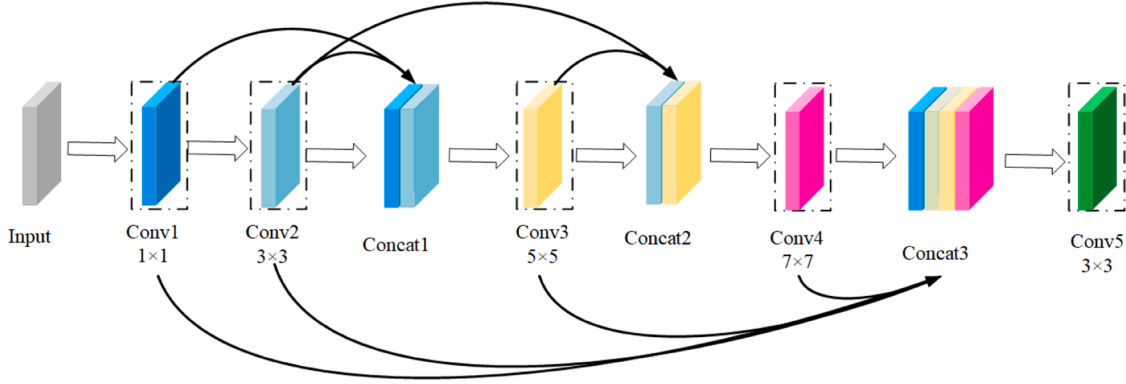


Fig. 3. AOD-Net network structure.

eliminates gaps between pixels in the feature layer and improves the information utilization rate. The design of the dilated rate in this paper meets the following conditions:

$$D = \max[D_{i+1} - 2l_i, D_{i+1} - 2(D_{i+1} - r_i), l_i] = \max[D_{i+1} - 2l_i, 2l_i - D_{i+1}, l_i] \quad (6)$$

Where, D_i signifies the maximum distance between two non-zero elements in layer i , and l_i represents the dilated rate. For the last layer of D_n , there are relationships $D_n = l_n$ and $D_2 \leq k$. At the same time, the common divisor between dilated rates $[l_1, l_2, \dots, l_n]$ cannot be greater than 1.

Based on the above analysis, this paper designs HDC with dilated rates [1-3] to replace the conv structure in the classical AOD-Net network, as depicted in Table 1.

The improved AOD-Net can capture more scale information and enhance the utilization rate of network space information and the overall performance. Table 1 shows that the improved AOD-Net comprises five convolution parts, among which Conv1 and Conv5 are standard convolution layers, and Conv2, Conv3, and Conv4 are comprised of HDC. After the hybrid dilated convolution structure is introduced, the receptive field of the AOD-Net network is enhanced from 15 to 75, which is five times higher than that of the original AOD-Net network.

Table 1
Different convolution structures of AOD-Net.

Conv Layers		Conv1	Conv2	Conv3	Conv4	Conv5
AOD-Net	kernel size	1 × 1	3 × 3	5 × 5	7 × 7	3 × 3
	dilated rate	1	1	1	1	1
	receptive field	1	3	7	13	15
	padding	0	1	2	3	1
	kernel size	1 × 1	3 × 3	5 × 5	7 × 7	3 × 3
Improved AOD-Net	dilated rate	1	[1,2,3]	[1,2,3]	[1,2,3]	1
	receptive field	1	13	37	73	75
	padding	0	[1,2,3]	[2,4,6]	[3,6,9]	1

3.2. Mixed attention fusion mechanism

Incorporating the attention mechanism into neural networks has been proven to focus on essential features of models and improve network performance. To address the issue of uneven concentration distribution in hazy images affecting the performance of the AOD-Net model, this article develops a new hybrid feature attention module found in Section 3.1, named ECPA, to further improve the effect of image dehazing.

3.2.1. Efficient channel attention module

Efficient channel attention (ECA) [36], as an efficient and lightweight network, can self-adaptively learn information about each channel in an image, as illustrated in Fig. 4. ECA adopts global average pooling to compress the input feature map, thereby integrating global spatial information and efficiently decreasing the number of parameters. Next, the 1×1 convolution is adopted to learn the importance of different channels. Finally, the feature map with the channel attention is fused with the initial input feature map to output the feature map with the channel attention.

ECA uses dynamic convolution self-adaptive learning to acquire the significance of various channels. The correlation between the magnitude of the convolution kernel and channels is as follows:

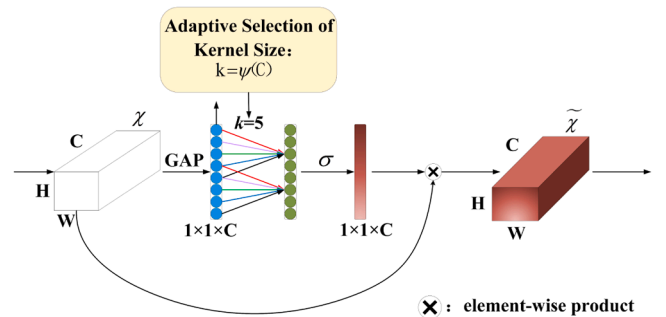


Fig. 4. ECA model. A lightweight network using adaptive kernel size selection and global average pooling.

$$k_{size} = \left\lceil \frac{\log_2(C)}{\gamma} + \frac{b}{\gamma_{odd}} \right\rceil \quad (7)$$

Where, *odd* stands for closest odd number, γ and b are set to 2 and 1 respectively to adjust the ratio of k_{size} to C . k_{size} is positively correlated with C , and a larger number of channels corresponds to a larger convolution kernel k_{size} , while a smaller number of channels corresponds to a smaller convolution kernel k_{size} . ECA overcomes the problem of feature map information extraction with different input sizes by the self-adaptive selection of convolution kernel sizes and completes cross-channel information interaction.

3.2.2. Pixel attention module

The distribution of haze concentration in ship images collected in practical applications could be more balanced. Therefore, the pixel attention (PA) [37] module is introduced in this paper to advance the feature extraction capacity of convolution and make the network focus more on the information features of pixel information and high-frequency image regions in heavy haze areas. The network structure of the PA module is depicted in Fig. 5.

In the PA module, when the feature map F is input, it passes through a convolution layer with a sigmoid activation function to attain the feature map F^* , as shown below:

$$F^* = \sigma(\text{Conv}(F)) \quad (8)$$

Where, σ is the sigmoid activation function.

Finally, multiply the input F and F^* element to obtain the final output F^{PA} :

$$F^{PA} = F \otimes F^* \quad (9)$$

3.2.3. Mixed attention module

Based on the above analysis, this paper integrates ECA and PA modules to design a mixed attention module (ECPA). The feature map after hybrid dilated convolution is captured by the ECA module, which assigns different weights to the channels, and the PA module assigns corresponding weights to the information of different pixels in the feature map. The designed ECPA module has greater flexibility in processing different channel information, pays more attention to haze and channel information characteristics, and effectively improves the effect of network dehazing. The ECPA module structure is depicted in Fig. 6.

3.3. Mixed loss function

A good loss function can make the network dehazing ability more accurate and achieve better results. Classical AOD-Net adopts the L2 loss function [21] for network training and assumes that the impact of noise on the image is independent of local features, which is inconsistent with the visual perception of human eyes. The structure similarity index measure (SSIM) [38] comprehensively considers image contrast, brightness, and structural similarity based on subjective perception, and the dehazing effect is more consistent with the actual scene. Building upon this, the present study designs a mixed loss function based on

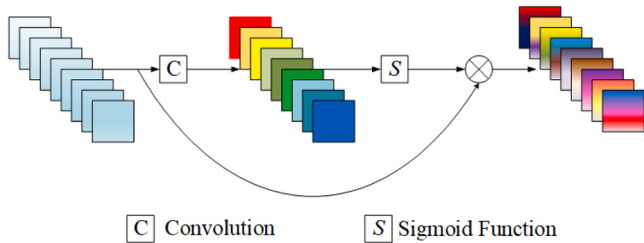


Fig. 5. PA model. It is composed of Conv and sigmoid function. The symbol “ $\otimes$ ” represents element-wise product.

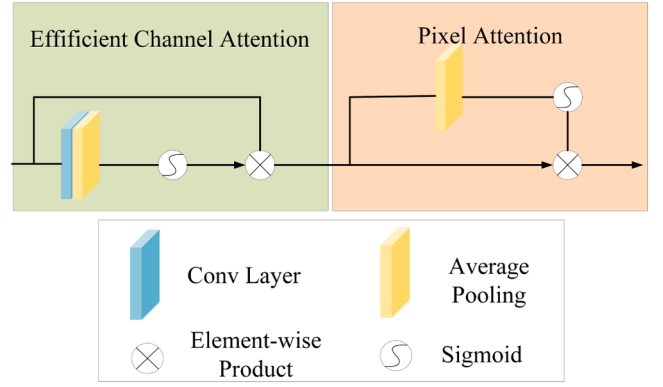


Fig. 6. ECPA module. It integrates ECA and PA modules to capture channel and pixel information in the feature maps.

SSIM. The mathematical expression of SSIM is as follows:

$$SSIM(x, y) = [L(x, y)]^a \cdot [C(x, y)]^b \cdot [S(x, y)]^c \quad (10)$$

$$L(x, y) = \frac{2\mu_x\mu_y + K_1}{\mu_x^2\mu_y^2 + K_1} \quad (11)$$

$$C(x, y) = \frac{2\sigma_x\sigma_y + K_2}{\sigma_x^2\sigma_y^2 + K_2} \quad (12)$$

$$S(x, y) = \frac{\sigma_{xy} + K_3}{\sigma_x\sigma_y + K_3} \quad (13)$$

Where, $L(x, y)$, $C(x, y)$, and $S(x, y)$ are image brightness, contrast, and structural similarity functions, respectively. a , b , and c are adopted to measure the relative importance of the three components, and $a = b = c = 1$; μ_x , μ_y , σ_x and σ_y are the mean and standard deviation of x and y , respectively; σ_{xy} is the covariance of x and y . K_1 , K_2 and K_3 are constants. Set $K_3 = K_2/2$ to prevent the denominator from being 0 and ensure the stability of the model.

However, the SSIM mentioned above is only applicable to specific dehazing models. Inspired by [38], this paper introduces multi-scale SSIM (MS-SSIM) into model training. MS-SSIM is the SSIM after multi-scale scaling of the image, as shown below:

$$MS_SSIM(x, y) = L(x, y)^{a_i} \cdot \prod_{i=1}^M [C(x, y)]^{b_i} [S(x, y)]^{c_i} \quad (14)$$

Similar to the definition of Formula (10), $a_i = b_i = c_i = 1$. M is the maximum scale of image scaling, and the loss function of $L^{MS-SSIM}$ is:

$$L^{MS-SSIM}(P) = 1 - MS_SSIM(\hat{P}) \quad (15)$$

Where, $\hat{P}$ is the center pixel of the pixel P .

In addition, considering that the L1 loss function can better retain the color and brightness in the image, it is introduced into the mixed loss function. The expression of the L1 loss function as follows:

$$L^1(P) = \frac{1}{N} \sum_{p \in P} |x(p) - y(p)| \quad (16)$$

To sum up, the mixed loss function designed in this paper is shown as follows:

$$L^{mix} = \beta \cdot L^{MS-SSIM} + (1 - \beta) G(\sigma_G^M) L^1 \quad (17)$$

Where, $\beta = 0.84$, and $G(\sigma_G^M)$ represents the Gaussian coefficients of σ_G on the scaling scale of M . As can be detected from the color and brightness of the original image are effectively preserved by using the mixing loss function, and the network dehazing effect is improved.

3.4. Improved AOD-Net model

Building upon the previous analysis, this paper mainly reconstructs the $K(x)$ estimation module in the AOD-Net model to improve the dehazing effect. The designed network structure is depicted in Fig. 7.

Specifically, firstly, to enhance the expression ability of traditional Conv for features of different scales, Conv2, Conv3, and Conv4 layers in the original $K(x)$ estimation module are replaced by HDC, which is named HDC1, HDC2, and HDC3, and expands the receptive field and further reduced information loss without advancing the amount of computation. In this process, the convolutional kernel size of HDC1 is set to 3, while HDC2 and HDC3 have convolutional kernel sizes of 5 and 7, respectively. Each HDC module has a stride of 1 and is composed of dilated convolutions with dilation rates of [1-3]. With this network architecture design, the receptive field of the output feature map is increased by 5 times, enabling the capture of more scale information and enhancing the utilization of spatial information in the network.

Considering the unequal distribution of haze concentration in the collected ship images, a mixed feature attention module ECPA is developed to advance the capability of detail information extraction in the channel and pixel dimensions. In the improved AOD-Net network structure, we apply the ECPA module to the output feature maps of each HDC, adaptively learning the weights of feature information at different levels to accommodate complex dehazing scenes. Additionally, we enhance the algorithm's feature fusion capability through residual

connections, addressing the issue of detail feature loss in AOD-Net. Finally, aiming at the phenomenon of image dimness and color distortion after dehazing, the mixing loss function L^{mix} is designed for network training, subsequently enhancing the dehazing performance of the model.

In summary, the improved AOD-Net network structure consists of 2 regular convolutions and 3 HDCs with ECPA. Similar to the AOD-Net network structure, Concat1 merges the information from Conv1 and HDC1, followed by further feature extraction through HDC2. Concat2 combines the feature information from HDC1 and HDC2 and then undergoes feature extraction through HDC3. Concat3 integrates multi-scale feature information from Conv1, HDC1, HDC2, and HDC3, and finally, outputs haze feature information through the Conv5 convolutional layer. By implementing these modifications, the reconstructed $K(x)$ estimation module in the AOD-Net model achieves improved dehazing effects. The devised network structure and associated modules contribute to enhanced feature extraction, quality of dehaze, and training strategy, collectively leading to superior dehazing performance compared to the original model.

To visually illustrate the effectiveness of HDC and ECPA modules, we print the channel feature map of the HDC layer and ECPA module. Fig. 8 shows that the HDC structure can catch more feature information. Through the use of dilated convolutions with varying kernel sizes, the HDC module expands the receptive field of Conv layers without increasing the computational burden. This allows the network to capture

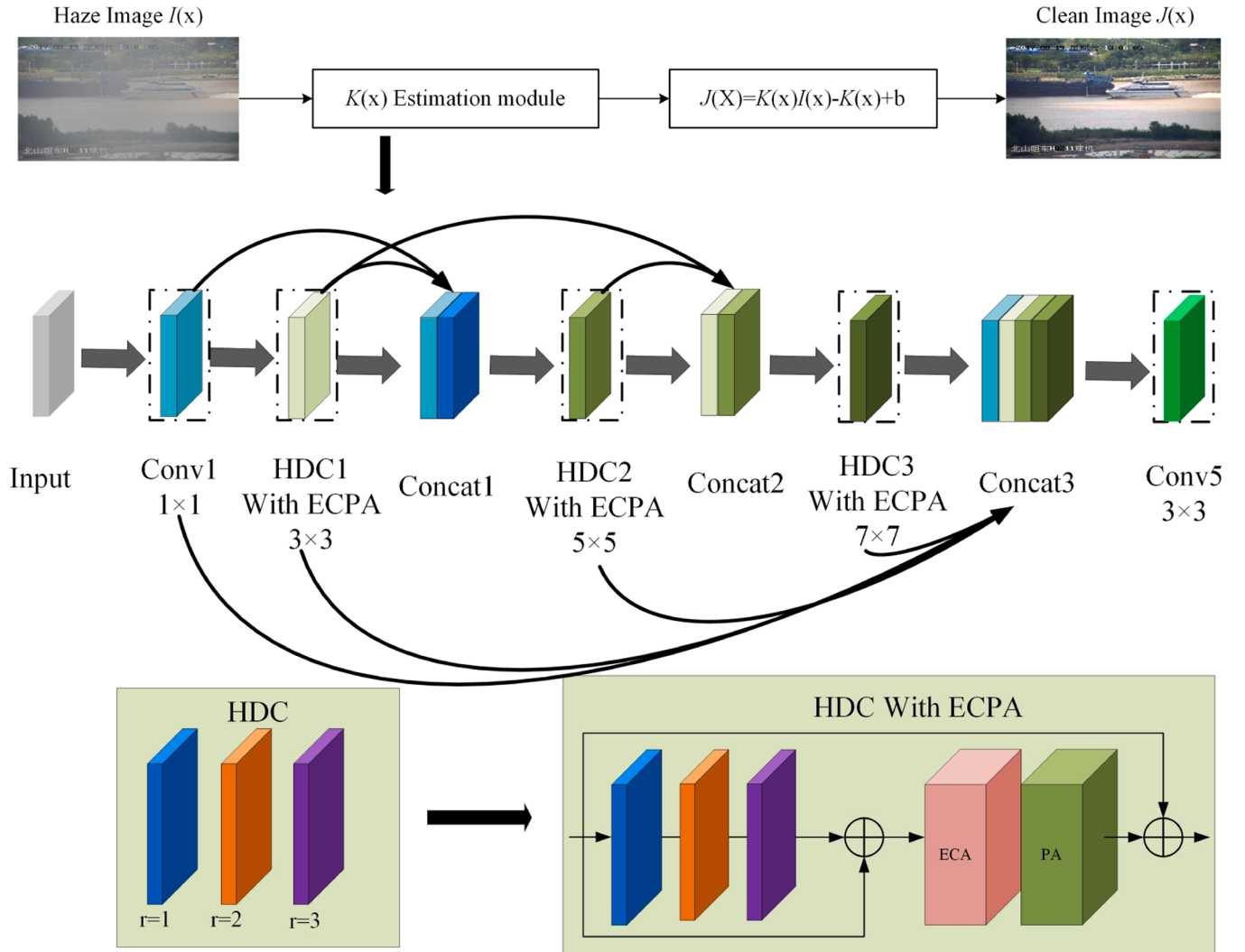
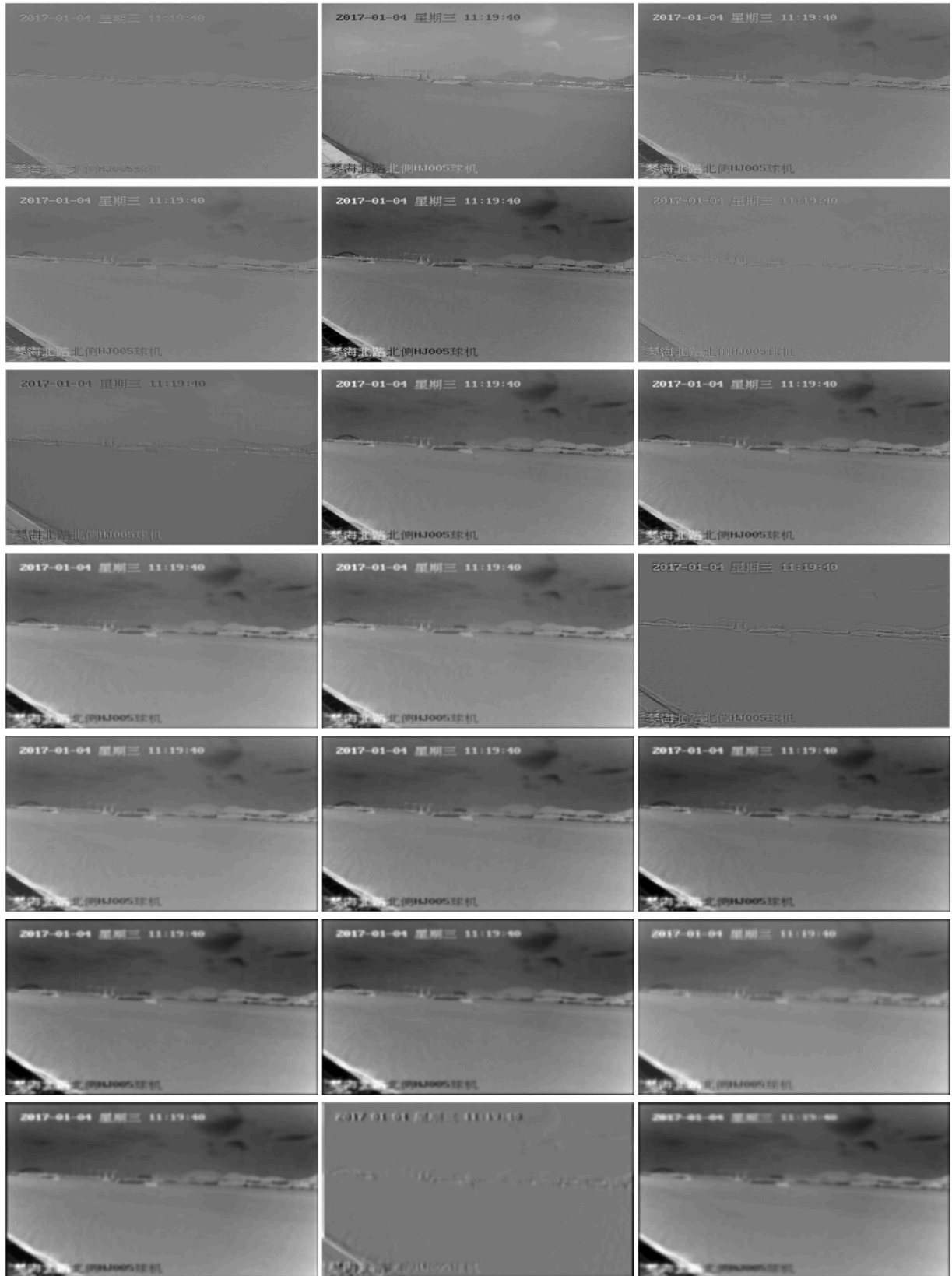


Fig. 7. Improved AOD-Net network structure.



(a) Channel 0

(b) Channel 1

(c) Channel 2

Fig. 8. Channel feature maps of each HDC module. From top to bottom, as the depth of the network increases gradually, the model can better extract high-level feature information.

features at multiple scales, enabling more comprehensive feature representation. The ECPA module plays a crucial role in enhancing haze removal while retaining image details. Fig. 9 shows that the ECPA improves the capability of extracting detailed information in both the channel and pixel dimensions and the different features adaptively learn different weights.

4. Experimental results and analysis

4.1. Datasets

In practice, obtaining hazy and clear images of the target scene under the same conditions is difficult, so it becomes a hot topic to use synthetic datasets for learning. Inspired by [39,40], based on ASM and clear images, this paper creates synthetic images with haze for model training.

In this paper, the public ship dataset (SeaShips) [41] is used as a dataset of clean images, this dataset currently consists of 7000 pictures with a resolution of 1920×1080 . Simultaneously, to meet a richer scene, the Huaxia industrial camera is used to collect the real river haze scene data, and the camera parameters are presented in Table 2. Then a total of 2925 pictures of ship images are gathered by camera with a resolution of 1280×720 . And 3800 ship images are randomly selected to synthesize haze images.

In addition, to improve the model's robustness, the above equipment is used in this paper to collect ship images of different haze concentrations under real-world in the Wuhan section of the Yangtze River. A total of 660 images are collected in the experiment for network training and testing, as shown in Fig. 10.

4.1.1. Synthetic haze dataset

According to ASM, the clear image $J_{\text{clean}}(x)$ can be synthesized into the hazy image $I_{\text{haze}}(x)$ by setting different atmospheric light values A and transmittance $t_{\text{trans}}(x)$. The details are as follows:

(1) Unified ship image resolution:

Considering that the acquired ship image and SeaShips data have different resolutions, the images are first processed as the same resolution:

$$J_{\text{reshape}}(1920, 1080) = f(J) \quad (18)$$

Where, J_{reshape} is the image size after unification, J is the original image resolution, and $f(\cdot)$ is the resolution unification function.

(2) Determine the image atomization size and atomization center:

This paper adopts the center point synthesis haze method for ship image hazing processing. The haze synthesis and diffusion are carried out through a center point of the hazy image. The greater the distance from the center point of the haze, the weaker the effect of haze synthesis.

$$\begin{cases} \text{size} = \sqrt{\max(\text{row}, \text{col})} \\ \text{center}(x, y) = (\text{row}/2, \text{col}/2) \end{cases} \quad (19)$$

Where, size is the image size and center is the atomization center.

Table 2

Camera parameters.

Parameter	Information
Power	18W
Horizontal running speed	$0^\circ - 220^\circ/\text{s}$
Horizontal rotation field	360°
Vertical running speed	$0^\circ - 90^\circ/\text{s}$
Vertical rotation field	$-95^\circ - 95^\circ$
Communication mode	RS485
Main stream resolution	1280×720
Baud rate	2400/4800/9600/19,200 bps

(3) Set atmospheric light value A and transmittance $t_{\text{trans}}(x)$;

Considering that the collected ship images are mainly composed of sky, water, and ship, $A = [0.5, 0.7]$ is set in the experiment. According to Eq. (2), transmittance $t_{\text{trans}}(x)$ is mainly determined by β . Set $\beta = [0.005, 0.035]$:

$$\begin{cases} A = 0.5 + 0.05 * s, & s = 0, 1, \dots, 4 \\ \beta = 0.005 + 0.005 * b, & b = 0, 1, \dots, 6 \end{cases} \quad (20)$$

(4) Composite ship image with haze:

By adjusting the values of A and transmittance β , ship images of different haze concentrations can be obtained. The partial samples of synthetic hazy images are displayed in Fig. 11.

4.2. Experimental detail

In this research, Pytorch with GPU is utilized to implement the enhanced AOD-Net model. Table 3 presents the hardware and software specifications employed in the study. The Adam optimizer is employed, and the training is carried out for 100 epochs with a batch size of 4. Both the initial learning rate and the attenuation coefficient are set to 0.0001.

4.3. Evaluation metrics

The evaluation metrics of dehazing in deep learning are the foundation for measuring the performance of dehazing results. The peak signal noise ratio (PSNR) [42], SSIM [43], CIEDE2000 [44], and Deep-Features [45] are employed to assess the effect of our algorithm on the synthetic haze data:

$$\text{PSNR} = 10 \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right) \quad (21)$$

$$\text{MSE} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (J(i, j) - I(i, j))^2 \quad (22)$$

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + K_1)(2\sigma_{xy} + K_2)}{(\mu_x^2 + \mu_y^2 + K_1)(\sigma_x^2 + \sigma_y^2 + K_2)} \quad (23)$$

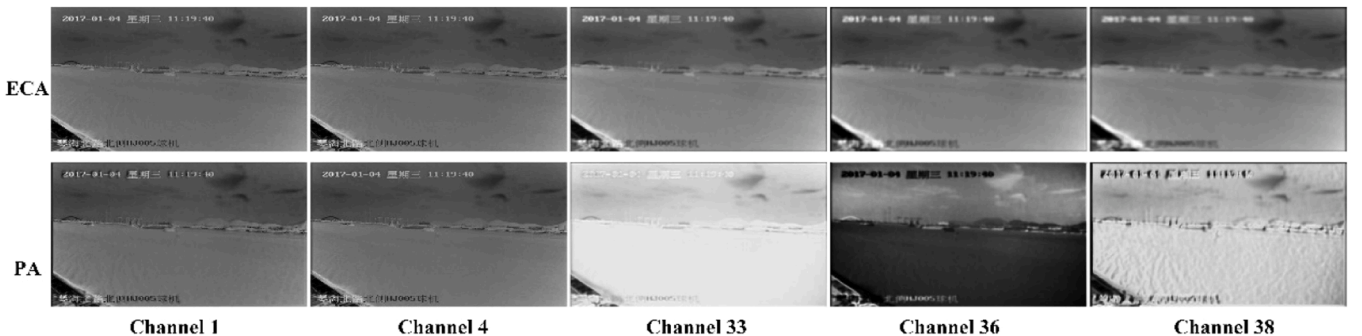


Fig. 9. Channel feature maps of ECPA module. For convenience, we show the feature maps of the preceding and following channels of the last ECPA module.

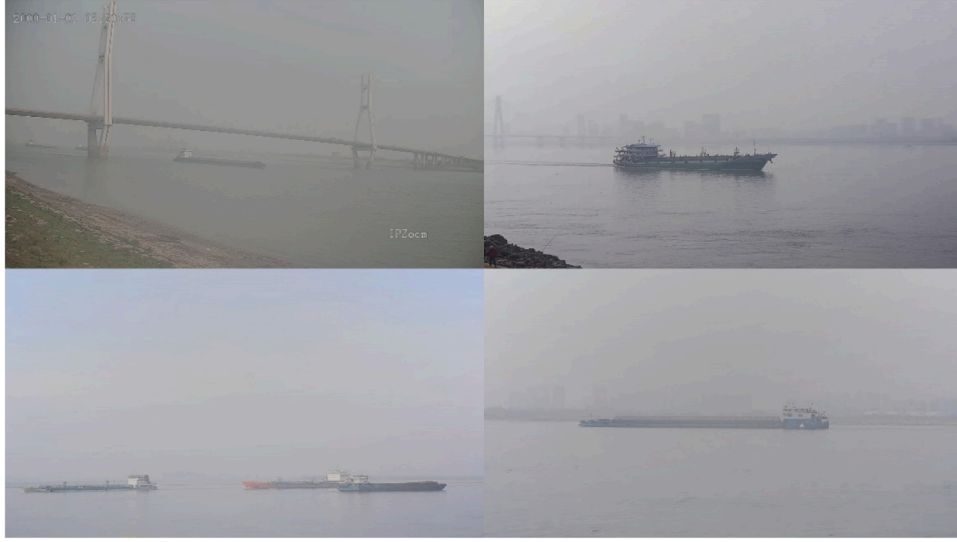


Fig. 10. Image of ships in the real-world haze.

Fig. 11. Composite haze ship image ($A = 0.5$, $\beta = [0.005, 0.035]$).

Table 3

Hardware and software environment.

Item	Type
CPU	11th Gen Intel(R) Core (TM)i5-11260H
Memory	16.0 GB
GPU	NVIDIA GeForce RTX 3050
Computer System	Windows 10
Deep Learning Framework	Pytorch 1.7.1

$$CIEDE2000 = \sqrt{\left(\frac{\Delta L'}{k_L S_L}\right)^2 + \left(\frac{\Delta C'}{k_C S_C}\right)^2 + \left(\frac{\Delta H'}{k_H S_H}\right)^2 + R_T \left(\frac{\Delta C'}{k_C S_C}\right)^2 \left(\frac{\Delta H'}{k_H S_H}\right)^2} \quad (24)$$

$$\text{cosine_similarity} = (F_{\text{haze}}, F_{\text{dehaze}}) = \frac{F_{\text{haze}} \cdot F_{\text{dehaze}}}{\|F_{\text{haze}}\| \times \|F_{\text{dehaze}}\|} \quad (25)$$

Where, MAX_I is the maximum pixel value in the image and $MAX_I = 255$, and MSE is the mean square error. It can be noticed that the larger the PSNR, the smaller the image distortion after image dehazing. The SSIM evaluation criteria is similar to human subjective perception, and its value is between $[0,1]$. The closer it gets to 1, the fewer differences the two images are. CIEDE2000 is used to evaluate the color restoration effect of images before and after dehazing. By comparing images before and after dehazing, CIEDE2000 can quantify the accuracy of color restoration. The smaller the color difference, the better the dehazing effect. $\Delta L'$, $\Delta C'$, and $\Delta H'$ represent the lightness difference, chroma dif-

ference, and hue difference between the two images, respectively. k_L , k_C , and k_H are scaling factors, usually set to 1. S_L , S_C , and S_H are weighting factors used to correct the uniformity of the color space. R_T is the rotation function used to compensate for the interaction between chroma and hue differences. And we use cosine similarity to evaluate the similarity of depth features between images. The closer this value is to 1, the higher the similarity between the two images in the depth feature space. Here, F_{haze} and F_{dehaze} are the feature vectors of hazy and haze-free images, respectively, and $(\cdot)$ denotes the dot product operation.

In addition, standard deviation (STD) [46] and information entropy (IE) [47] are adopted to evaluate the dehazing outcome of the model on real-world hazy ship images. Among them, STD describes the dispersion degree of pixels in an image with their average value. The larger the value, the better the visual effect. As an indicator of the amount of information in an image, the larger the value of IE, the more information in the image. The expression of the two is:

$$STD = \sqrt{\frac{1}{W \cdot H} \sum_{i=1}^W \sum_{j=1}^H (p_{ij} - \mu)^2} \quad (26)$$

$$IE = - \sum_{i=1}^N P(i) \log_2 P(i) \quad (27)$$

Where, W and H are the width and height of the image respectively; p_{ij} is the gray value corresponding to pixel (i,j) ; μ is the average pixel value of the whole image; P_i is the probability of gray value i .

4.4. Experimental results and analysis

To validate the performance and effectiveness, the most representative and high-performing learning models, covering DCP [14], Light-DehazeNet [48], GCA-Net [49], FFA-Net [24], and AOD-Net [21], are utilized to perform a comprehensive comparison.

4.4.1. Results on synthetic datasets

The suggested algorithm is evaluated on the synthetic dataset, three separate scene hazy images are utilized to show the visual comparisons and the model dehazing results on different models are displayed in Fig. 12. It is clear that DCP appears to have a noticeable color loss phenomenon in the restored image. This is because the ship image contains many water surface and sky areas, which causes a large deviation in the estimation of atmospheric light value by DCP, affects the estimation accuracy of transmittance, and finally leads to the edge effect. The images recovered by Light-DehazeNet, GCA-Net, and AOD-Net are generally dark. Among them, AOD-Net still has a large amount of haze in the dehazing result, and its processing effect of haze in the image could be better, leading to some details being obscured. Light-

DehazeNet and GCA-Net have relatively little haze residue but slight color distortion. Both FFA-Net and our method can effectively dehazing in the image. In contrast, our method has a clearer outline and better visual effects after image dehazing.

To further quantify the performance of different approaches, the PSNR and SSIM on the test set are calculated in the synthetic haze ship dataset. The experiment outcomes are put forth in Fig. 13. We know that our method also demonstrates strong advantages in quantitative metric analysis, with PSNR and SSIM values generally higher than other models. FFA-Net and our model exhibit comparable distributions at the 25th percentile and median. Combined with Fig. 12, it is evident that our model generally has a strong capability to remove varying degrees of haze, with a relatively concentrated data distribution. Since the data simulated different levels of haze, the optimized model demonstrates a strong capability in handling light haze, resulting in a more elongated data distribution. This makes the overall data distribution of PSNR and SSIM slightly better than that of FFA-Net, further indicating that our method achieves better overall dehazing effects and more stable performance. And from Table 4 we can see that ours gets an average PSNR of 22.352 dB, which outperforms all other comparison methods.



Fig. 12. Visual dehazing results on synthetic datasets.

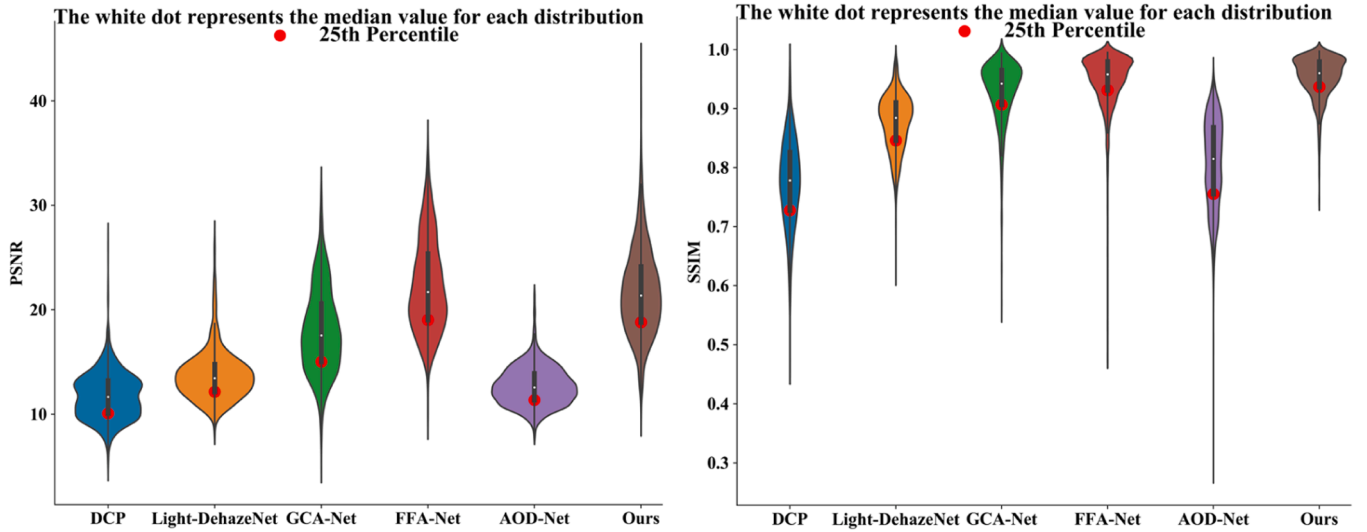


Fig. 13. PSNR and SSIM on synthetic dataset.

Table 4
Quantitative comparison on different models.

Metrics	DCP	Light-DehazeNet	GCA-Net	FFA-Net	AOD-Net	Ours
PSNR (dB)	11.774	13.752	18.027	21.703	12.699	22.352
SSIM	0.775	0.876	0.928	0.943	0.807	0.949
CIEDE2000	25.963	19.839	14.516	6.207	23.566	4.274
cosine_similarity	0.887	0.900	0.907	0.980	0.942	0.983

Regarding SSIM, our method gets high results, which is 0.949, and the proposed method is effective in different types of dehazing tasks through these performance improvements.

To verify the statistical significance of the results, the T-test is employed to compare the PSNR and SSIM values between FFA-Net and our model. The results indicate that our method significantly outperforms FFA-Net in both PSNR and SSIM ($P_{PSNR} = 1.92^{-6} < 0.05$,

$P_{SSIM} = 2.9^{-4} < 0.05$). According to the just noticeable difference (JND) standard, a PSNR difference of over 0.5 dB and an SSIM difference of over 0.01 are noticeable, further proving the superiority of our method in dehazing performance. Additionally, considering the CIEDE2000 and cosine similarity metrics, our algorithm achieved top rankings with values of 4.274 and 0.983, respectively. This indicates that the images processed with our method have smaller color differences and higher similarity, resulting in reliable visual enhancement effects that align more closely with human subjective perception.

4.4.2. Results on real-world datasets

To evaluate the generalizability of this algorithm, experiments are executed on real-world datasets, and the result is depicted in Fig. 14. A severe overexposed effect occurs in DCP results, where certain areas of the image are too bright or too dark. Light-DehazeNet can remove a large amount of haze, but it has a poor treatment effect on the haze in the sky and water surface area, presenting a fuzzy phenomenon. Although GCA-Net can properly deal with the haze at the edge of the ship target, a large amount of haze remains in other areas, and severe color distortion occurs. AOD-Net has a normal dehazing effect, weak processing ability for details, and dark brightness. These models exhibit high sensitivity to environmental conditions or image features, making it challenging to effectively remove extremely dense haze in complex natural conditions. FFA-Net and our method show strong dehazing ability and can obtain relatively clear images. Nonetheless, the output of our method is more natural, and more details are retained, which can properly deal with the haze of different concentrations.

Due to the lack of corresponding clear images in the real-world haze scene, STD and IE indicators are used in this paper to quantify the dehazing capabilities of various approaches, as shown in Table 5. When using our method, the STD reaches 37.754, which is better than other comparative methods, further indicating that the method we put forth in this study has a better visual effect on dehazing. In IE, all comparison methods are relatively even. Our method can retain more image information and is competent for dehazing under different real-world haze conditions.

4.5. Real-time analysis of the model

To study the dehazing efficiency of various models, the average handling time of all models on the verification set is statistically analyzed, as shown in Table 6. Different network topologies also show significant differences in dehazing efficiency. AOD-Net has the highest dehazing efficiency, and the handling time of each image is only 0.18 s. The efficiency of FFA-Net in the image dehazing method shown is low, and the handling time of each image is 3.26 s, which cannot meet the real-time requirements. DCP, Light-DehazeNet, GCA-Net, and our method all have better efficiency in dehazing. The method put forth in this study redesigns the network structure and adds the ECPA module. The average processing time is 0.93 s, which can satisfy the demands of real-time dehazing treatment in real-world applications.

4.6. Ablation study

To verify the ability of the HDC module, ECPA module, and mixed loss function to improve network performance, an ablation experiment is conducted on the improvement of the three modules in the same experimental environment as above. The test results of the synthetic dataset can be seen in Table 7. From the results of the ablation study, we know that the dehazing performance of our model improves gradually with the increase of each module. Specifically, since the large receptive

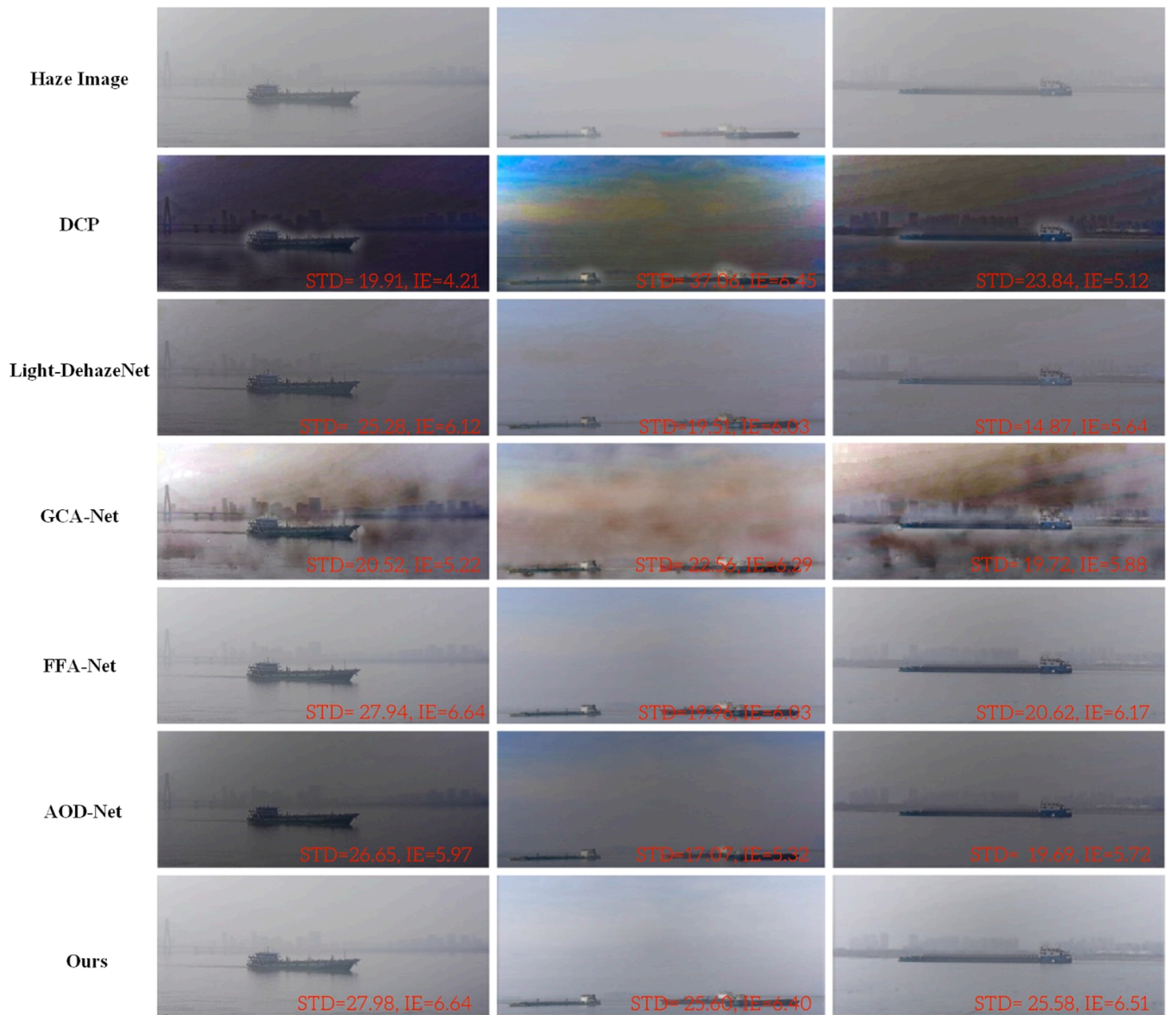


Fig. 14. Visual dehazing results on real-world datasets.

Table 5
Quantitative comparison on different models.

Metrics	DCP	Light-DehazeNet	GCA-Net	FFA-Net	AOD-Net	Ours
STD	33.323	25.089	31.717	37.411	34.967	37.754
IE	6.708	6.327	6.655	6.798	6.685	6.917

Table 6
Comparison of average per-image running time on different models.

Models	DCP	Light-DehazeNet	GCA-Net	FFA-Net	AOD-Net	Ours
Time(s)	0.63	0.20	0.81	3.26	0.18	0.93

field, HDC obtains the first improvement rank on both PSNR and SSIM, with an average gain of 7.723 and 0.12, which demonstrates the effectiveness of hybrid dilation convolution. Besides, the ECPA hybrid attention mechanism and the mixed loss function are also beneficial for the dehazing performance.

Table 7
Ablation experiments of the proposed model.

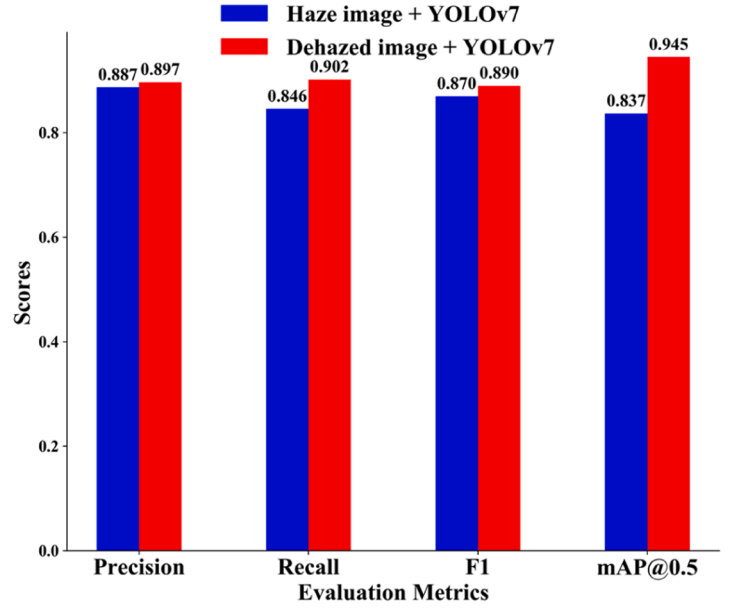
Models	HDC	ECPA	Mixed loss	PSNR(dB)	SSIM
AOD-Net	/	/	/	12.699	0.807
	✓	/	/	20.422	0.927
	✓	✓	/	21.860	0.936
	✓	✓	✓	22.352	0.949

4.7. Effectiveness of the proposed model for object detection in a hazy environment

A surveillance system's performance can be significantly reduced due to uncertain weather conditions such as storms, fog, or haze. Most maritime surveillance systems use object detection techniques to detect suspicious activities in a specific water area. To tackle the vision problem in the hazy atmosphere, the maritime surveillance system must be provided with a visibility enhancement algorithm. With this in mind, we evaluate the proposed method's effectiveness for object detection problems in hazy environments. Fig. 15 demonstrates the visual



(a) Vision detect results



(b) Evaluation metrics

Fig. 15. Performance evaluation of the object detection in the hazy environment with and without image dehazing process. a. The detection results on haze and dehazing images. b. The evaluation metrics of detection results.

detection results obtained from the experiments using the YOLOv7 [50] with our method. In Fig. 15(a), it can be seen that when using a visual object detector to detect a single ship, good performance is exhibited in both hazy and clear environments. However, when there are multiple ships, the detector fails to effectively detect all targets, resulting in false positives and missed detections. This is due to the decreased quality of visual images in hazy conditions, which prevents the detector from accurately extracting target features. After applying our dehazing preprocessing to the visual images, the detection results significantly improve, accurately extracting moving ship targets and greatly avoiding situations of missed detection and false detection. To further quantify the effectiveness of object detection [51], we use a series of evaluation metrics to measure the performance of the visual detector. In Fig. 15(b), using our method to process hazy images effectively improves the accuracy of ship detection. After dehazing the ship image, the mAP increases by 10.8 %, Recall increases by 5.6 %, and the other two metrics also show improvements.

To sum up, by enhancing image clarity and visibility, our method facilitates more accurate and reliable ship detection and navigation, contributing to improved safety and efficiency in maritime operations. By addressing these challenges and achieving significant improvements in dehazing quality, our research contributes to the development of more effective and robust dehazing techniques that can be applied in various applications, including intelligent ship navigation and visual perception enhancement.

5. Discussion

Image dehazing plays an important role in various real-world applications such as intelligent surveillance, remote sensing, and autonomous driving. By reducing the effects of haze, visibility is improved, which enhances the quality and reliability of subsequent advanced computer vision tasks. With the rise of deep learning, convolutional neural networks (CNN) can effectively learn and handle various environmental challenges, including issues caused by haze, such as illumination changes and visual degradation [52,53]. This indicates that while traditional dehazing techniques still hold significance, deep learning has

made it possible for some applications to potentially bypass explicit dehazing steps. However, real-world environmental conditions are constantly changing. Particularly in the context of intelligent ship development, the key to safe navigation in severe weather conditions lies in effectively perceiving the surrounding environment and making timely and accurate maneuvers. Existing deep learning methods cannot fully address all visual perception issues and may encounter problems such as loss of feature information or misjudgment. Therefore, the importance of using explicit dehazing techniques provides a direction for future research, exploring the combination of the end-to-end learning capabilities of deep learning with the advantages of traditional dehazing methods to achieve more efficient and accurate image processing techniques.

6. Conclusions and future work

In hazy weather, due to the existence of a large number of small particles suspended in the air and other strongly scattering media, the video or image acquired by the vision system usually has low visibility, poor contrast, and other phenomena, which affects the progress of other advanced computer vision tasks. The inland navigation environment is complex and changeable, which is easily affected by the scattering of haze particles, which could be more conducive to intelligent monitoring of water traffic and identifying and tracking surface targets. To restore the visual scene degraded by haze, many scholars have proposed a series of image-dehazing methods in recent years. Existing methods are generally based on the ASM model to obtain clear images by estimating information such as atmospheric light and scene depth. However, the robustness and generalization performance of the model needs to be further improved, and the dehazing effect is not ideal for the real natural environment. To overcome the above problems and guarantee the navigation safety of intelligent ships in hazy weather conditions, this paper proposes a real-time image dehazing method built upon the AOD-Net model for complex haze environments in inland rivers. We first introduce HDC to enhance the expression ability of convolution operation for different feature information. Compared with the classical Conv, the receptive field in the new network structure is expanded by 5 times,

which effectively reduces the loss of feature information, and does not bring the computation surge. Then, the ECPA module is added to the network structure to improve the ability of the model to remove haze of different concentrations. Experimental results also show that the module can retain more edge information and local features. In addition, we use the mixed loss function to conduct the final network training, which can effectively solve the color distortion or dim problems existing in other models. Specifically, to advance the representation ability of characteristics in the convolution process, HDC is used to reconstruct the dehazing network structure, effectively improving the range of feature receptive field without increasing the amount of computation. Relative to the classical Conv, the receptive field in the new network structure is expanded five times, effectively limiting the feature information loss. Secondly, considering that the collected images of inland waterway ships contain a large number of sky and water surface areas, and the haze concentration in different areas of the images also varies greatly, a hybrid attention module ECPA is proposed to be designed by fusing the channel and the pixel attention mechanism. By giving distinct weights to the light haze and heavy haze regions in the image, the model can improve its dehazing ability while retaining more detailed features. Finally, to enhance the image restoration quality, the mixing loss function is designed for network training. To validate the effectiveness and superiority of the introduced dehazing method, the usual dehazing methods are selected to conduct comparative tests on the synthetic haze and real-world haze datasets, respectively. The experimental data demonstrate that the dehazing method recommended in this article has a more realistic visual effect on the synthetic haze dataset, with PSNR reaching 22.352 and SSIM reaching 0.949, both of which are the highest among all comparison methods. Additionally, the CIEDE2000 and feature similarity metrics both achieved top rankings, effectively realizing reliable visual enhancement effects that align more closely with human visual perception. At the same time, our method also shows a decisive advantage in the real-world haze dataset, with STD reaching 37.754 and IE 6.917, which further indicates that our dehazing method is more effective than other approaches and can attain more image details.

The dehazing algorithm introduced in this article can attain a relatively ideal haze reduction result under different hazy scenarios, and obtain clearer and natural ship images. From the visual detection result, we know that our method facilitates more accurate and reliable ship detection and navigation, contributing to improved safety and efficiency in maritime operations. Enhanced visual perception enables intelligent ships to better detect and recognize obstacles, navigational markers, and other ships in hazy conditions. Our method contributes to improved safety, efficiency, and risk reduction in maritime transportation. This improves the overall reliability and robustness of maritime transportation networks, even in challenging environmental conditions, provides technical support for intelligent supervision of inland water and safe navigation of ships, and further promotes the development of intelligent shipping.

Although our method has achieved excellent performance in ship image dehazing, there are still some limitations in this work. Firstly, considering that our model will be used for shipborne computer equipment, the constructed network structure is relatively simple. As we all know, deeper networks can mine more scale information of images, so we may lose some ability to extract feature information. In future work, we will design a more efficient network with a certain depth to solve this problem. Secondly, for the restoration results of real haze scenes in inland rivers, our method is difficult to avoid some noise and a small amount of haze. Finally, our method is to experiment with static conditions. Although we have achieved good results in the comparative experiment, it cannot meet the task of video dehazing in practical application, and the speed of image dehazing needs to be further improved. In future work, we will focus on solving the above problems, and expand the real haze dataset to explore a more robust dehaze model.

CRedit authorship contribution statement

Yuanzhou Zheng: Writing – review & editing, Validation, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Long Qian:** Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Yuanfeng Zhang:** Supervision, Methodology, Conceptualization. **Jingxin Cao:** Visualization, Formal analysis. **Xinyu Liu:** Visualization, Formal analysis. **Yong Ma:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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